Department of Biotechnology, School of Energy Technology

B.Tech Biotechnology

Course Matrix for Semester – II (Batch 2024-2028)

Sem.	Category Code	Course Code	Course Name	Th	Tut	Pra	Hrs	Cr
II	HSC	24HS105T	Professional Communication	2	0	0	2	2
	BSC	24MA104T	Mathematics for Biotechnology – II	3	1	0	4	4
	ESC	24CP102T	Computer Programming – II	1	0	0	1	1
	ESC	24CP102P	Computer Programming Laboratory – II	0	0	2	2	1
	BSC	24CH101T	Engineering Chemistry	3	0	0	3	3
	BSC	24CH101P	Engineering Chemistry Laboratory	0	0	2	2	1
	ESC	24EE101T	Elements of Electrical and Electronics Engineering	3	0	0	3	3
	ESC	24EE101P	Elements of Electrical and Electronics Engineering Laboratory	0	0	2	2	1
	HSC	24HS103T	Indian Knowledge System	2	0	0	2	2
	HSC	24YOG101	Yoga, Health & Hygiene	0	0	2	2	1
		24NSS101	National Service Scheme (NSS)					
		24NCC101	National Cadet Corps (NCC)					
	HSC	24HS104T	Organizational Behaviour	1	0	0	1	1

Syllabus for the Courses of Semester – II (Batch 2024-2028)

24HS105T					Professional Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES

1. To strengthen the communication skills of professionals to make them ready for the modern workplace.
2. To fine tune their professional skills and expertise using communication skills.
3. To participate in the life-long learning process with confidence and certainty.
4. To develop analytical, research, and organizational skills through communication skills for a fulfilling career.

UNIT 1 TECHNICAL WRITING **07 HRS.**

- E-mails
- Report Writing
- Proposal writing (Project, Research and Business)
- Review Writing (Books and Articles)

UNIT II: ORATORY SKILLS **07 Hrs.**

- Art of Introducing oneself
- Presentation Skills
- Public speaking

UNIT III: EMPLOYABILITY SKILLS **07 Hrs.**

- Deciphering JD (Job Descriptions)
- Cover Letter Drafting
- Resume Writing
- Group Discussion
- Interview Skills

UNIT IV: ADVANCED READING AND CRITICAL THINKING **07 Hrs.**

- Critical Thinking
- Language and Communication:
- Ethical Considerations:
- Interdisciplinary Perspective:
- Communication Skills:
- Philosophy of Science

TOTAL: 28 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Develop the technical writing skills essential for effective communication in professional settings, including business environments, academic institutions, and research organizations.

CO2 : Master the art of public speaking by delivering clear, engaging, and persuasive presentations to diverse audiences, demonstrating confidence, poise, and effective communication techniques.

CO3 : Demonstrate the ability to decipher job descriptions, extracting key requirements, and qualifications to align their skills and experiences effectively.

CO4 : Effectively prepare for and participate in group discussions, demonstrating active listening, critical thinking, and the ability to articulate ideas coherently and persuasively.

CO5 : Understand the importance of clear, honest communication in engineering practice and ethical decision-making in upholding privacy rights and societal well-being.

CO6 : Explore interdisciplinary approaches to problem-solving.

TEXT/REFERENCE BOOKS

1. Effective Communication Skills. Kul Bhushan Kumar, Khanna Book Publishing, 2022.
2. Remedial English Grammar. F.T. Wood. Macmillan.2007
3. Sharma, Sangeeta and Binod Mishra. Communication Skills for Engineers and Scientists. New Delhi: PHI Learning Pvt. Ltd., 2009.
4. Kaul, Asha. Business Communication. Delhi: Prentice-Hall of India, 2006.

24MA104T					Mathematics for Biotechnology – II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To make students acquainted with the basics of functions of several variables.
2. To be able to formulate and solve various biotechnology problems using the methods of solving ODEs.
3. To study the use for double and triple integral to find area and volume.
4. To be able to evaluate problems related to probability and distribution.

UNIT I: FUNCTIONS OF SEVERAL VARIABLES **10 Hrs.**

Functions of two variables, Limits and Continuity, Partial derivatives, Total derivatives, Maxima and minima, Lagrange multipliers method.

UNIT II: MULTIPLE INTEGRALS **12 Hrs.**

Definition Evaluation of double integral (Cartesian – Polar form), Change of orders, Change of variables, Evaluation of triple integral, change of variables (Cartesian to spherical – and cylindrical), Applications, Area, Volume.

UNIT III: ORDINARY DIFFERENTIAL EQUATIONS **10 Hrs.**

Differential equations of first order and first degree, Exact differential equations, Integrating factors, Higher order differential equations with constant coefficients, Rules for finding C.F. and P.I., Cauchy and Legendre's linear equations, Applications of higher order differential equations in solving biotechnology problems.

10 Hrs.

UNIT IV: PROBABILITY AND DISTRIBUTION

Sample Space and Events; Axioms, Interpretations and Properties of Probability; Expectation; conditional Probability; Total probability, Bayes' Rule, Random variables; Measures of central tendency and dispersion.

Total: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand the basic concepts of multivariable calculus, and differential equations.
CO2 : Demonstrate the use of multivariable calculus, differential equations.
CO3 : Apply appropriate tool/method to extract the solutions of biotechnology problems.
CO4 : Analyze the obtained solution in context with theory.
CO5 : Evaluate area, volume using multiple integrals.
CO6 : Create a mathematical model of biotechnology interest.

TEXT/REFERENCE BOOKS

1. B. S. Grewal, Elementary Engineering Mathematics, Khanna Publisher.
2. M. D. Raisinghania, Ordinary and partial differential equations, S Chand Publication.
3. S. C. Gupta and V. K. Kapoor, Fundamental of Mathematical Statistics, S. Chand and Sons
4. Sheldon M. Ross, Introduction to probability and statistics for engineers and scientists, Academic Press publication.

24CP102T					Computer Programming – II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
1	0	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

1. To impart the basic concepts of Python Programming language
2. To be familiar with data structures available in Python.
3. To understand the concept of function & object-oriented programming.
4. To use the file and exception handling for designing applications

UNIT 1 INTRODUCTION AND LANGAUGE FUNDAMENTALS **4 Hrs.**

Definition and its historical background, Comparison with other programming languages. Python IDEs **Tokens:** Identifiers, Keyword **Data Types:** Int, Float, Bool, Complex, Str, List, Tuple, Dictionary and Set, Type conversion, Concept of mutability, immutability and reusability **Operators:** Arithmetic, Relational, Logical, Identity and Membership, Input and Output methods Control Flow Conditional statements: If, elif and else, nesting, Iterative or loop statement: for, while, for else and while else, Transfer statements: break, continue, pass.

UNIT 2 FUNCTIONS **3 Hrs.**

Function Declaration and calling, Types of Function arguments, scope of variable, Recursive function Recursive function **Types of function:** Anonymous or lambda, Map, Filter, Reduce, Function aliasing, Nested function, Decorator function, Decorator chaining Generator function

UNIT 3 OBJECT ORIENTED PROGRAMMING **4 Hrs.**

Class, Object, Object reference, Constructor and self-variable, Types of variables, Types of method, Destructor, Composition, Aggregation, Inheritance, Polymorphism, Abstract classes.

UNIT 4 FILE AND EXCEPTION HANDLING **3 Hrs.**

File handling Types of files, writing and reading with text file, important functions of file handling, working with directories, CSV file handling, pickling and unpickling **Exception Handling** exceptions handling using try-except blocks. Raising exceptions and custom exception classes.

14 Hrs.

COURSE OUTCOMES

CO1- Understand the basic concepts of programming with python.
CO2- Demonstrate proficiency in using different data types, operators, and control structures in Python programs.
CO3- Demonstrate the usage of both built-in and user-defined functions
CO4- Define and use classes in Python to represent attributes and methods, and demonstrate composition, aggregation, inheritance and polymorphism.
CO5- Implement exception handling mechanisms to deal with errors.
CO6- Demonstrate file operations and exception handling.

TEXT/REFERENCE BOOKS

1. John V Guttag, Introduction to Computation and Programming Using Python, Prentice Hall of India.
2. Allen Downey, Jeffrey Elkner and Chris Meyers, How to think like a Computer Scientist, Learning with Python, Green Tea Press.
3. Al Sweigart, Automate the Boring Stuff with Python
4. Martin C. Brown, Python: The Complete Reference, Osborne, McGraw-Hill
5. R. Nageswara Rao, Core Python Programming, Dreamtech Press

24CP102P					Computer Programming Laboratory – II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To impart the basic concepts of Python programming language
2. To be familiar with data structures available in Python.
3. To understand the concept of function & object-oriented programming.
4. To use file and exception handling for designing applications

List of Experiments:

1. **Language fundamentals**:
   Understanding interpretation, program execution flow, tokens, variables and keywords and identifiers, constants and variables
2. **Input Output Operations:**
   Basic data types, mutability, immutability, Operators, Input /output statements, format specifiers, escape sequences
3. **Decision making, Looping control structures**
   If, elif and else, nesting, Iterative or loop statement: for, while, for else and while else, Transfer statements: break,
4. **Advanced datatypes**
   String, List, Tuples, Dictionary, Set
5. **Functions:**
   Basics of functions, types of arguments
6. **Types of functions**
   Anonymous function, function aliasing, nested function, decorator function and generator function
7. **Object Oriented programming**
   class, object, types of variable and methods, composition, aggregation, inheritance, types of inheritance
8. **File Handling:**
   writing and reading with text file and csv file, working with directories, pickling and unpickling

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1- Apply the concepts of variables, decision making and looping with python.
CO2- Apply various data structures available in Python to solve computational problems.
CO3- Implement advanced function concepts such as decorator chaining and generator functions.
CO4- Design and implement object-oriented solutions to programming problems using composition, aggregation, inheritance and polymorphism.
CO5- Develop Python scripts, for file handling tasks.
CO6- Create and manage files in python.

TEXT/REFERENCE BOOKS

1. John V Guttag. “Introduction to Computation and Programming Using Python”, Prentice Hall of India.
2. Allen Downey, Jeffrey Elkner and Chris Meyers "How to think like a Computer Scientist, Learning with Python", Green Tea Press.
3. Al Sweigart, “Automate the Boring Stuff with Python”
4. Martin C. Brown, “Python: The Complete Reference, Osborne, McGraw-Hill
5. R. Nageswara Rao, “Core Python Programming”, Dreamtech Press

24CH101T					Engineering Chemistry					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To develop the fundamental understanding about traditional materials.
2. To provide the knowledge about structural features, synthesis, properties of various categories of advanced materials.
3. To develop the knowledge and skills for different characterization techniques of materials.
4. To provide the knowledge about the role of chemistry in modern engineering applications.

UNIT I: CHEMISTRY OF ENGINEERING MATERIALS **12 Hrs.**

Traditional Materials: Introduction and classification of materials; metallic materials, polymeric, ceramic materials Advanced Materials: Introduction to nanomaterials: Properties and application; Carbonaceous materials (fullerene, carbon nanotube, graphene, etc.); Composite materials; Liquid crystals: Classification and Application

UNIT II: MODERN ANALYTICAL TECHNIQUES **10 Hrs.**

Instrumentation, principle and characterization of materials: X-ray diffraction (XRD), Electro analytical techniques; FTIR, UV-visible spectroscopy; Thermal analysis (TGA-DTA-DSC); Chromatographic techniques (GC, HPLC)

UNIT III: ADSORPTION, CATALYSIS AND KINETICS **10 Hrs.**

Adsorption - Characteristics, Classification, Application , Adsorption isotherms- Freundlich, Langmuir & BET Chemical Kinetics - Rate law, Arrhenius equation, Transition state theory, Collision theory; Complex reactions Catalysis - Homogeneous and Heterogeneous Catalysis; Mechanism of Catalysis; Industrial Applications of catalysts

UNIT IV: CHEMISTRY OF ENERGY DEVICES **10 Hrs.**

Principles of primary and secondary batteries, Fuel Cells and their operation principles, Principles and uses of supercapacitors; Photocatalytic hydrogen production: Principles and challenges; Traditional and new generation solar cells.

TOTAL : 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : **Remember** the structural features and properties of different classes of traditional materials
CO2 : **Classify** advanced materials like nanomaterials, carbonaceous and composite materials, and liquid crystals
CO3 : **Apply** the skills by understanding various instrumental techniques for characterisation of materials.
CO4 : **Analyze** the key concepts in engineering chemistry viz. adsorption and chemical kinetics and laterally ponder over the applications of such concepts in engineering challenges.
CO5 : **Justify** the important insights into the industrial application of different types of catalysis via analysing mechanisms of catalysis.
CO6 : **Develop** the knowledge on the role of chemistry in various modern engineering applications such as in energy devices.

TEXT/REFERENCE BOOKS

1. W.D. Callister, **"An Introduction to Materials Science & Engineering",** John Wiley & Sons (2007).
2. MW Barsoum, **"Fundamental of Ceramics",** IOP publishing (2003).
3. T. Pradeep, "Text book of Nanoscience and Nanotechnology", Mc. Graw Hill Education (2003).
4. Murty, Shankar, B Raj, Rath, Murday, **"Textbook of Nanoscience and Nanotechnology",** Springer (2013).
5. V. Raghavan, **"Materials Science and Engineering",** Prentice-Hall of India Private Limited (2003).
6. A. Douglas, Donald Skoog, M.West**, "Principles of Instrumental Analysis",** Cengage, 6th Edition, (2014)
7. Jain & Jain, **"Engineering Chemistry"** by Dhanpat Rai Publishing Company

24CH101P					Engineering Chemistry Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To enhance and develop scientific and analytical skills
2. To relate concepts learned in chemistry and engineering to the real-world situations.
3. To acquire skills to perform laboratory experiments.
4. To demonstrate safe and proper use of standard chemistry glassware and equipment.

LIST OF EXPERIMENTS

1 **Iodometry**– To determine the strength of given copper sulphate solution by titrating against N/20 sodium thiosulphate (hypo) solution

2 **Iodimetry**– To determine the strength of given ascorbic acid by titrating against standard N/10 iodine solution

3 **Complexometric Titration**– To determine the total, permanent and temporary hardness of given water by complexometric titration using standard 0.01M EDTA solution

4 **pH metric titration**– To determine the strength of given HCl solution using a standard NaOH solution by performing a pH-metric titration

5 **Conductometric titration**– To determine the strength of given HCl solution using a standard NaOH solution by performing a conductometric titration

6 **Chemical Kinetics**– To study the kinetics of decomposition of sodium thiosulphate by a mineral acid

7 **Drawing chemical structures** - To Draw Chemical Structures of organic molecules using ChemDraw

8 **Colorimetric determination**: To determine the concentration of copper present in the effluent of electroplating industries by using colorimeter.

9 **Detection of biomolecule**: Detection of the presence of carbohydrates in test solution by using Benedict's reagent

10 **Preparation of drug molecule**: Preparation of Aspirin from salicylic acid

11 **Polymerization**– To prepare a polymer (Nylon 6,10), identify the functional groups by FT-IR

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recall the concepts learned in chemistry and engineering to the real-world situations.
CO2 : Show the ability to identify, analyse and interpret the results from the experiments.
CO3 : Experiment with the instrumental method using conductometer and pH meter.
CO4 : Analyse compounds by titrimetric, gravimetric and instrumental methods.
CO5 : Determine the concentration of unknown solutions by spectrophotometric method.
CO6 : Predict the reaction rate and predict the order and rate constant.

TEXT/REFERENCE BOOKS

1. V. K. Ahluwalia, S Dhingra, A. Gulati, College Practical Chemistry, Universities Press.
2. J.B. Baruah, P Gogoi, Foundations of Experimental Chemistry, PharmaMed Press.
3. S. S. Sawhney, M. S. Jassal, S.P. Mittal, A Text Book of Chemistry Practical, Vol. I & II, APH Publishing Corp.

24EE101T					Elements of Electrical and Electronics Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart knowledge on DC and AC circuits.
2. To learn construction, working principle and characteristics of transformer and induction machines.
3. To introduce students to various means for electrical safety, protection of electrical installations and Batteries
4. To understand rectification through p-n junction diode, applications of diode and Transistor Characteristics

UNIT I: BASICS OF ELECTRONICS **10 Hrs**

Semiconductor Diodes and Applications, Diode as a Switch, Diode as a Rectifier, Half Wave and Full Wave Rectifiers with and without Filters; Breakdown Mechanisms, Zener Diode – Operation and Applications; Opto-Electronic Devices–LEDs, Photo Diode and Applications; Silicon Controlled Rectifier (SCR) – Operation, Construction, Characteristics, Ratings, Applications.

Transistor Characteristics covering, Bipolar Junction Transistor (BJT) – Construction, Operation, Amplifying Action, Common Base, Common Emitter and Common Collector Configurations, Field Effect Transistor (FET) – Construction, Characteristics of Junction FET.

UNIT II: DC CIRCUITS **10 Hrs**

Electrical circuit elements (R, L and C), voltage and current sources, dependent and independent sources, Ohms Law, temperature co-efficient of resistance, Kirchhoff current and voltage laws, voltage and current divider circuit, Thevenin and Norton Theorems and their equivalents, maximum power transfer and superposition theorems, nodal and mesh analysis, star-delta transformation, Time domain analysis/natural response of first order RL and RC Circuit.

UNIT III: AC CIRCUITS **10 Hrs**

Generation of AC voltage, representation of sinusoidal waveforms, RMS values of different sinusoidal waveforms, Rectangular and Polar representation of phasor, Sinusoid representation in time and frequency domain, Analysis of single-phase ac series circuits consisting of R, L, C, RL, RC, RLC combinations, instantaneous power, average power and reactive power, complex power and power factor.AC parallel circuit and its solution in admittance form, resonance in AC series and parallel circuits, polyphase circuits, star and delta representation of polyphase circuits, power measurement in polyphase circuits.

UNIT IV: INTRODUCTION TO ELECTRICAL MACHINES AND ELECTRICAL INSTALLATION **12 Hrs**

Magnetic materials and its B-H characteristic, Faraday's Law of Electromagnetic Induction; **Single Phase Transformers:** Construction, working principle, types, EMF equation, ideal transformer, practical transformer, Operation of transformer on no-load and load, **Induction Machine:** construction, types of 3-phase induction motors, working principle, production of rotating magnetic field, operation, starting and running torques, Torque-slip characteristics, Power Stages in an induction motor.

Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Types of Wires and Cables, Earthing, Types of Batteries, Important Characteristics for Batteries. Elementary calculations for energy consumption, Electrical safety rules.

TOTAL : 42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Demonstrate application of different diodes in circuits and Understand the Transistor characteristics

CO2 – Understand electrical circuits using network theorems.

CO3 – Compare the behaviour of R, L and C and their combinations in AC circuits.

CO4 – Analyze balanced polyphase systems in star and delta configurations

CO5 – Understand the construction, working and basic characteristics of transformer and induction machines

CO6 -- Recognize the importance of protective devices, electrical safety measures and characteristics of Batteries

TEXT/REFERENCE BOOKS

1. J. Bird, **"Electrical Circuit Theory and Technology"**, Routledge, Tailor and Francis Group
2. D. P. Kothari and I. J. Nagrath, **"Basic Electrical Engineering"**, Tata McGraw Hill
3. B. L. Theraja, **"Electrical Technology",** Vol. 1, S. Chand Publication, New Delhi.
4. Surjit Singh, **"Electrical Estimating and Costing"**, Dhanpat Rai and Co.
5. Boylestad and Nashlesky, "**Electronic Devices and Circuit Theory**", PHI

24EE101P					Elements of Electrical and Electronics Engineering Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To gain practical knowledge on DC and AC circuits
2. To learn operation of electrical instruments and electrical machines
3. To introduce students to various means for electrical safety, protection of electrical installations
4. To understand characteristics and applications of semiconductor diodes and transistors.

LIST OF EXPERIMENTS

1. Introduction to elements of electrical engineering laboratory and to study different electrical measuring instruments.
2. To Verify Ohm's law with linear resistors and find power dissipation in resistor.
3. To validate Thevenin's and Norton theorem for DC circuits.
4. To validate Superposition and Maximum Power Transfer theorem for DC circuits.
5. To evaluate the AC R, L and R-L series circuit performance and to measure the active power, the reactive power and the apparent power connected with single phase AC supply.
6. To evaluate performance of AC R-L parallel circuit and to measure the active power, the Reactive power and the apparent power connected with single phase AC supply.
7. To perform a direct load test on a single-phase transformer and determine the efficiency and voltage regulation at different loads.
8. To measure three phase power using two wattmeter method.
9. To draw the time-current characteristics of MCB and study the function of ELCB.
10. To obtain the VI characteristic of silicon and germanium diodes.
11. To obtain reverse characteristics of Zener diode.
12. To study half wave, full wave and bridge rectifiers.
13. To determine the DC Characteristics of BJT in CE Configuration.
14. To study the types of batteries and their characteristics

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – To understand the basic operation of electrical equipment's & measuring instruments.
CO2 – To perform various network theorems for DC circuits.
CO3 – To understand the performance of AC circuit with R, L load.
CO4 – To evaluate star and delta configuration of polyphase system and measure power in polyphase system.
CO5 – To evaluate the performance of single phase transformer.
CO6 – To understand the working principle of semiconductor diodes, transistor characteristics and its applications.

24HS103T					Indian Knowledge System					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
1	0	0	1	1	25	50	25	--	--	100

COURSE OBJECTIVES

1. To connect with the tradition of IKS
2. The students will be able to define Philosophical foundations of IKS.
3. To understand Fundamentals of Art and Architecture, Traditional and Historical Town Planning & Ancient Indian Art & Architecture.

UNIT 1 INTRODUCTION TO IKS **07 Hrs.**

- What is IKS? Historical and philosophical foundations
- Knowledge Framework and classifications in IKS
- Indian scheme of knowledge
- The knowledge triangle
- Framework for establishing valid knowledge
- Deductive or inductive logic framework
- Potential fallacies in the reasoning process

UNIT II: IKS AND ENGINEERING DISCIPLINES **07 Hrs.**

- Engineering marvels of ancient India
- Civil engineering: Urban planning, water management
- Materials engineering: Sustainable materials, traditional construction techniques Material science and Metallurgy in IKS
- Mechanical engineering: Robotics and automation in ancient India
- The role of IKS in addressing climate change challenge
- IKS principles for resource management and conservation

UNIT III: IKS AND MATHEMATICS **07 Hrs.**

- IKS and Indian Mathematics, Algebra, Geometry, Trignometry, Binary Mathematics, Magic
- IKS and Indian Astronomy, Pañcāṅga – The Indian calendar system
- Astronomical Instruments -Yantras

UNIT IV: IKS AND OTHER DISCIPLINES **07 Hrs.**

- Indian Administration
- Concept of state,
- Anushashan Parwa of Mahabharat
- Kautilya's Arthshastra
- Social and Political Philosophies of Ancient India
- IKS and Medicine: Ayurveda - A Holistic Approach to Health and Wellbeing

TOTAL: 28 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Define the fundamental concepts of Indian Knowledge System
CO2 : Describe various cost concepts of IKS relevant from engineering perspective.
CO3 : To Apply Mathematics in the Vedas and Śulva SūtraSystems employed for representing numbers Spherical trigonometry &Celestial Sphere
CO4 : To Analyse the science of Astronomy and the different units of time discussed in the texts
CO5 : To be able to weigh the IKS systems vis s vis the current applications in various spheres of understanding
CO6 : To design applications and processes by incorporating thebl traditional knowledge to the present day concerns

TEXT/REFERENCE BOOKS

1. Introduction To Indian Knowledge System: Concepts and Applications by by B. Mahadevan, Nagendra Pavana, Vinayak Rajat Bhat.
2. The Vedas by Sri Chandrasekharendra Saraswati/Sankaracharya of Kanchi Kamakoti Peetham (Author)
3. Indian Knowledge Systems – Vol 1 & 2 by Avadhesh K. Singh, Kapil Kapoor

24YOG101					Yoga, Health & Hygiene					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To impart the students with the basic concepts of physical education, sports, and yoga for health and wellness.
2. To familiarize the students with health-related exercises, sports, and yoga for overall growth and development.
3. To create a foundation for professionals in physical education, sports, and yoga.
4. To impart the basic knowledge and skills to teach physical education, sports, and yoga activities.

ACTIVITY I: KINESIOLOGY AND CARDIO FITNESS TEST

Introduction to Kinesiology and the Physiological Basis of Conditioning, Sports Psychology, and the Cooer Cardio Fitness Test 12-Minutes Run/Walk: How to Start Walking/Runing: Get expert tips, tools, and training.

ACTIVITY II: YOGA

Introduction to Yogasana and Yoga Therapy: A Rehabilitation Tool and the Effect of Yoga on Exercise Endurance as Assessed by Cardiorespiratory Efficiency Tests: Studt on yogic practices that promote and improve respiratory and cardiovascular function and enhance physical fitness.

ACTIVITY III:GAMES AND SPORTS LEAGUE

Practice sessions for outdoor and indoor games, event-wise practice, and team games organized on the sport, game-wise practice as per the student's interest: Football, Chess, Cricket, Tennis, Basketball, Volleyball, Athletics (Relay), Pickleball.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Discover the balance of health and happiness through the basic principles and practices of physical education, sports, and yoga.
CO2 : Understand that the students will be able to be instructed on physical activities, sports, yoga practices, theories, and rules of various games for healthy living.
CO3 : Analyze emerging trends and issues in world sports and develop leadership qualities among students to conduct, organize, and officiate physical education, sports, and yoga events at schools, colleges, and the community.
CO4 : Practice on the field and in the indoor yoga hall.
CO5 : Develop a spirit of teamwork and fair play.
CO6 : Demonstrate understanding by participating in games and sports leagues.

TEXT/REFERENCE BOOKS

1. Athletic Track and Court Marking Handbook of Games And Sports – Rajesh Agola.
2. Asana, Pranayama, and Kriyas - Swami Satyanand Swami.Munger.
3. Sports Games and Rule, Regulation - Pankaj Vinayak Pathak
4. Yogic Prakriyanche Margdarshan – Dr.M.L.Gharote - (The Lonavala Yoga Research Institute,Lonavala)

24NSS101					National Service Scheme (NSS)					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To develop a sense of civic and social responcibility.
2. To identify the needs and problems of the community and involve them in problem-solving.
3. To engage in creative and constructive social action.
4. To develop social character and leadership through NSS.

ACTIVITY I: ENVIRONMENT AND SUSTAINABILITY -WATER - WAST MANAGEMENT

Volunteering work for Environment & Sustainability (water and waste management) and Tree Plantation.

ACTIVITY II: NSS 7 DAYS SPECIAL CAMP

Volunteering for tree planting, agriculture compost, tree guard,Gujarat Skill Development Mission, and social activities in the village as per the government NSS manual.

ACTIVITY III: FIT INDIA MISSION

Volunteering for Cardio Fitness, Yoga, Running, Mission Olympics, Self-Defense, and Agneepath Mission.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Identify the needs and problems of the community.
CO2 : Understand the importance of his / her responsibilities towards society.
CO3 : Analyze the environmental and societal problems/issues.
CO4 : Evaluate the existing system and propose practical solutions for the sustainable development.
CO5 : Develop a government or self-driven projects effectively in the field.
CO6 : Understand the government or self-driven projects effectively in the society.

TEXT/REFERENCE BOOKS

1. NSS Course Manual, Published by NSS Unit,PDEU (https://www.pdpu.ac.in/nssreport.html)
2. Government of Gujarat NSS Cell (https://nss.gov.in/gujarat-1)
3. Government of India NSS Cell, Activities reports and manual (https://nss.gov.in/)

24NCC101					National Cadet Corps (NCC)					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To develop discipline, character, brotherhood, the spirit of adventure and ideals of selfless service amongst young students
2. To develop youth leadership qualities in the students.
3. To induce social consciousness among students through various NCC camps

ACTIVITY I: INTRODUCTION TO NCC

Introduction to NCC, aims and objectives, structure and organization of NCC, cardinals of NCC, NCC Flag, oath of NCC, NCC Song, incentives of NCC

ACTIVITY II: NATIONAL INTEGRATION

Importance of national integration and awareness, necessity, national interests, objectives, threats and opportunities, unity in diversity

ACTIVITY III: NCC CAMPS, SOCIAL SERVICE, AWARENESS AND COMMUNITY DEVELOPMENT ACTIVITIES

Social awareness & community development, health & hygiene, environment awareness and conservation, cadets will participate in various activities e.g., blood donation camp, swachhata abhiyan, constitution day, etc., participation into NCC camps like ATC, CATC, NIC, COC, TSC, RDC, leadership camps, etc.

ACTIVITY IV: DRILL, WEAPON TRAINING AND ADVENTURE ACTIVITIES

Types of drill, foot drill, general and words of command, saluting, weapon training, map reading, field craft & battle craft, Introduction to infantry weapons & equipment, obstacle and weapon training (during camps), adventure training, participation into Republic and Independence day ceremonial parades at university.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Know about the history of NCC, its organization, and incentives of NCC for their career prospects.
CO2 : Understand the importance of Nation building and individual contribution to the same
CO3 : Maintain discipline and team spirit
CO4 : Build the character and leadership qualities
CO5 : Understand that drill as the foundation for discipline and to command a group for common goal.
CO6 : Develop the sense of self-less social service for better social & community life.

TEXT/REFERENCE BOOKS

2. Cadet's Handbook SD/SW- Common Subjects, all wings by DG NCC, New Delhi

24HS104T					Organizational Behaviour					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE	
1	0	0	1	1	25	50	25	--	--	100

COURSE OBJECTIVES

1. To enable the development of a perspective to diagnose & efficiently deal with matters of human behaviour in organisations
2. To enrich understanding of the dynamics of interaction and integration between the individual and the organization.
3. To explore the organization system in entirety to capture the full complexity of organizational behaviour.
4. Critically appraise the potential effects of important developments in the external environment on organizational behaviour

UNIT 1 INTRODUCTION TO ORGANIZATIONAL BEHAVIOUR **04 HRS.**

Definition and scope of organization, behaviour, and organizational behaviour, Historical developments and emerging concerns in OB, Perception in organizational behaviour: elements, principles, and biases, Learning theories, conditioning, application in organizations, Personality theories, Attitudes in the workplace.

UNIT II: EMOTIONS, STRESS MANAGEMENT, GROUP DYNAMICS AND INTERPERSONAL RELATIONSHIPS **04 Hrs.**

Understanding emotions in organizational behaviour, Emotional labor, Stress management techniques in organizational settings, Group formation and dynamics in organizations, Group decision-making techniques and their application, Team building strategies and effective communication, Influence, power dynamics, and organizational politics

UNIT III: LEADERSHIP AND ORGANIZATIONAL CULTURE **03 Hrs.**

Leadership theories: overview and application in organizational settings, Leadership styles and their impact on organizational behaviour, Understanding organizational culture and climate

UNIT IV: ORGANIZATIONAL CHANGE AND STRUCTURE, BALANCING WORK AND LIFE **03 Hrs.**

Organizational change processes, Resistance management during organizational change, Organizational structure and design, Job satisfaction: determinants and impact on employee behaviour, Stress prevention and management techniques, Work-life balance strategies and their role in organizational behaviour

TOTAL: 14 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Critically gage the potential effects of important developments in the external environment on organizational behaviour.
CO2 : Analyze organizational behavioural issues in the context of organizational behaviour theories, models, and concepts.
CO3 : Analyze the behaviour of individuals and groups in organizations in terms of the crucial factors that effect organization behaviour.
CO4 : Analyze organizational behavioural matters in the context of organizational behaviour theories, models, and concepts.
CO5 : Demonstrate how the organizational behaviour can integrate in understanding the motivation (why) behind behaviour of people in the organization.
CO6 : Manage conflict in organizational context and deal with stress.

TEXT/REFERENCE BOOKS

1. Fiona M. Wilson, Organizational Behaviour and Work, Oxford University Press, 5th Edition, 2018.
2. Stephen P. Robbins, Timothy A. Judge, Neharika Vohra, Organization Behaviour, Pearson Education, 18th Edition, 2019.
3. Ashwathappa, K., Organizational Behaviour, Himalaya Publication, 12th Edition 2019.
4. wathappa, K., Organizational Behaviour, Himalaya Publication, 12th Edition 2019.